Vaccine Hesitancy Detection in Multilingual Social Media Comments using Traditional Machine Learning, Deep Learning and Transformer Models

Introduction

Vaccine hesitancy has emerged as a major global public health challenge, especially on social media platforms where misinformation spreads rapidly through user-generated content. The World Health Organization (WHO) has identified vaccine hesitancy as one of the significant threats to global health. Social media discussions surrounding vaccines often contain misinformation, distrust, fear, political narratives, and mixed public opinions, making automatic detection of vaccine-related stance an important Natural Language Processing (NLP) task.

This project focuses on multilingual vaccine hesitancy detection using social media comments collected from YouTube. The original objective of the project was to build a multimodal framework combining text, image, network, temporal, and geospatial modalities for detecting vaccine-related misinformation and stance in code-mixed Indian social media discussions. However, practical challenges related to data availability, annotation feasibility, and project time constraints resulted in refinement of the project scope.

The final implementation focused primarily on multilingual text classification and temporal analysis. The dataset consisted of English, Hindi, and code-mixed YouTube comments manually annotated into three stance categories:

- Pro-vaccine
- Anti-vaccine
- Neutral

Three different modeling approaches were explored:

1. TF-IDF + LinearSVC baseline model
2. Convolutional Neural Network (CNN)
3. Multilingual BERT (mBERT)

Additionally, temporal analysis was performed using the published timestamps of comments to study stance trends over time.

The objective of the project was not only to achieve high classification accuracy, but also to understand the practical challenges involved in multilingual social media stance detection, annotation consistency, modality feasibility, and model behavior under limited and noisy real-world datasets.

Literature Review

Traditional NLP and Machine Learning Approaches

Traditional Natural Language Processing methods for text classification commonly rely on sparse vector representations such as Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF). TF-IDF assigns higher importance to words that are informative within a document while reducing the importance of frequently occurring generic words.

Support Vector Machines (SVMs) have historically performed well on text classification tasks due to their effectiveness in handling high-dimensional sparse feature spaces. LinearSVC, a linear implementation of SVM, is particularly popular for sentiment analysis and stance detection because of its computational efficiency and strong performance on limited datasets.

However, traditional machine learning models rely heavily on handcrafted features and often struggle to capture semantic context, sarcasm, multilingual variations, and code-mixed language patterns commonly found in social media text.

Deep Learning for NLP

Deep learning approaches introduced distributed word representations and contextual feature learning into NLP tasks. Neural networks can automatically learn meaningful patterns from data without extensive feature engineering.

Convolutional Neural Networks (CNNs) became popular for sentence classification after the work of Kim (2014), where convolution filters were used to capture local semantic patterns and n-gram features from text embeddings. CNNs are computationally efficient and often perform well on short-text classification tasks such as tweets, comments, and reviews.

CNN-based models are especially useful when dealing with noisy social media text because they can capture localized phrase-level information better than sparse statistical methods.

Transformer-based Models

Transformer architectures significantly changed NLP research by introducing self-attention mechanisms capable of learning contextual relationships between words. BERT (Bidirectional Encoder Representations from Transformers) introduced bidirectional contextual understanding and achieved state-of-the-art performance on numerous NLP tasks.

Multilingual BERT (mBERT) extends this concept to multiple languages using shared multilingual embeddings. mBERT can process multilingual and code-mixed text without requiring separate language-specific models.

Transformer-based models generally outperform traditional models when large-scale annotated datasets and computational resources are available. However, transformer models may overfit or underperform on small datasets due to their high parameter complexity.

Challenges in Multilingual Social Media NLP

Social media text presents several unique challenges:

- Informal writing style
- Spelling variations
- Code-mixed language usage
- Emojis and slang
- Sarcasm and irony
- Context-dependent meaning
- Extremely noisy data

Indian social media discussions frequently contain English-Hindi code-mixed content, transliterated words, and inconsistent grammar. These characteristics increase annotation difficulty and reduce model separability between stance categories.

Furthermore, stance classification differs from sentiment analysis because stance often depends on contextual interpretation rather than direct emotional polarity.

Methodology

Initial Project Scope

The original project objective involved building a multimodal vaccine hesitancy detection framework integrating:

- Text modality
- Image modality
- Network analysis
- Geospatial analysis
- Temporal analysis

However, during implementation several feasibility challenges emerged.

Image Modality Challenges

The project initially intended to combine images with social media comments. However, practical data collection limitations made this difficult:

- YouTube comments do not provide paired images for individual comments.
- Video thumbnails were available, but the same thumbnail could not reliably represent all comments under a video.
- Reddit vaccine discussions rarely contained associated images.
- Reliable image-comment alignment was not feasible within the project timeframe.

Therefore, image-based multimodal fusion was excluded from the final implementation.

Geospatial Modality Challenges

Geospatial analysis was initially considered using geotagged social media information. However:

- YouTube comments rarely contain geographic metadata.
- Reddit discussions also lacked usable geolocation information.
- Reliable user location extraction was not feasible.

As a result, geospatial analysis was not included in the final system.

Temporal Modality Selection

Unlike image and geospatial metadata, temporal metadata was consistently available through comment timestamps (`published_utc`). Therefore, temporal analysis was selected as the additional modality explored in the project.

Although multimodal fusion was initially planned, obtaining semantically aligned image-text pairs for individual comments was not feasible within the Indian social media context. YouTube comments do not provide paired images per comment, while Reddit vaccine discussions rarely contained associated visual content. Using unrelated vaccine images alongside textual comments would effectively create parallel unimodal pipelines rather than a truly aligned multimodal representation. Therefore, the final implementation focused on modalities that maintained direct association with the collected textual data.

Network-based misinformation analysis was also considered during the initial project design. However, meaningful network modeling typically requires large-scale interaction graphs, reply propagation structures, or influencer relationship data. The limited availability of relevant vaccine-related discussions within the targeted Indian demographic, combined with project time constraints, reduced the feasibility of implementing reliable network analysis within the current study.

The final project scope focused on:

- Multilingual text classification
- Temporal analysis of stance evolution

Dataset Collection

Data collection was performed primarily using the YouTube Data API v3.

The study was intentionally restricted to COVID-19 vaccine discussions within the Indian social media context. Vaccine hesitancy patterns can vary significantly across vaccine types due to differences in public perception, rollout timelines, emergency authorization, and social discourse. COVID-19 vaccines generated uniquely polarized online discussions compared to more established vaccines such as rabies or childhood immunization programs. Restricting the dataset to COVID-19 discussions improved thematic consistency within the analysis.

Multiple YouTube videos related to vaccination discussions were identified and comments were scraped using API endpoints. Additional Reddit data was also scraped during the project, although it was not fully integrated into the final experiments due to time limitations.

Approximately 3,500 unlabeled YouTube comments were collected.

Due to annotation time constraints, a manually labeled subset of 320 comments was created for model training and evaluation.

The final dataset contained:

- English comments
- Hindi comments
- English-Hindi code-mixed comments

The multilingual and code-mixed nature of the dataset increased the realism of the experiments while also increasing classification difficulty.

Annotation Process

Initial Four-Class Annotation

The first annotation scheme consisted of four classes:

- Pro-vaccine
- Anti-vaccine
- Hesitant
- Irrelevant

However, during experimentation several issues emerged:

- Significant overlap existed between hesitant and neutral statements.
- Social media comments were highly noisy and ambiguous.
- The models struggled to separate semantically similar classes.
- Annotation consistency became difficult.

The initial four-class configuration produced lower separability between labels.

Transition to Three-Class Annotation

To improve label consistency and class separability, the annotation strategy was refined.

The final three-class configuration used:

- Pro
- Anti
- Neutral

Highly irrelevant comments were removed and the annotation guidelines were made stricter and more text-focused.

An important realization during annotation was that human annotation often considered the broader video context, while the machine learning models only received the comment text itself.

This mismatch created inconsistencies between annotation reasoning and model input.

After simplifying the labels and making the annotation criteria more classification-oriented, the models showed improved F1 scores and better class separability.

Text Preprocessing

Several preprocessing steps were applied:

- Lowercasing
- URL removal
- Mention removal
- Whitespace normalization

Initially, preprocessing removed all non-English characters using regular expressions. However, this negatively affected multilingual comments because Hindi and code-mixed text information was removed.

The preprocessing pipeline was later modified to preserve multilingual characters:

```
text = re.sub(r'[^\w\s]', ' ', text)
```

This modification improved classification performance, indicating that multilingual tokens carried important stance-related information.

TF-IDF + LinearSVC Baseline

The baseline model used:

- Word-level TF-IDF features
- Character-level TF-IDF features
- LinearSVC classifier

Feature extraction included:

- Word n-grams (1,2)
- Character n-grams (3,5)

Character-level features were included to better capture spelling variations and code-mixed patterns common in social media comments.

Class balancing was handled using:

```
class_weight='balanced'
```

This model served as the classical machine learning baseline for comparison against deep learning approaches.

CNN Model

A Convolutional Neural Network (CNN) was implemented for text classification.

The CNN architecture consisted of:

- Embedding layer
- Multiple convolution filters
- Max pooling layers
- Dropout regularization
- Fully connected output layer

The CNN used multiple kernel sizes to capture local semantic patterns from the text.

Advantages of CNN for this project included:

- Better contextual representation than TF-IDF
- Ability to learn semantic phrase patterns
- Robustness to noisy social media text
- Effective performance on short text classification

Early stopping was used to reduce overfitting.

Multilingual BERT (mBERT)

The transformer-based approach used:

`bert-base-multilingual-cased`

mBERT was selected because the dataset contained multilingual and code-mixed comments.

The model was fine-tuned using supervised learning with manually labeled comments.

Weighted cross-entropy loss was used to reduce class imbalance effects.

Experiments with different learning rates and epoch settings were conducted.

It was observed that:

- Longer training caused overfitting on the small dataset.
- Three epochs provided better generalization than five epochs.

Temporal Analysis

Temporal analysis was performed using the `published_utc` timestamps associated with each comment.

Initially, monthly aggregation was attempted. However, because the dataset contained only 320 labeled comments distributed across multiple years, monthly visualizations became highly sparse and statistically unstable.

For example:

- Some months contained only one or two comments.
- Proportion plots fluctuated excessively.
- Small sample sizes created misleading temporal trends.

Therefore, quarterly aggregation was adopted instead.

Quarterly aggregation:

- Reduced noise
- Improved visualization readability
- Produced more stable temporal trends
- Better suited the dataset size

Temporal analysis included:

- Quarterly label counts
- Stacked bar visualizations
- Temporal heatmaps
- Peak quarter analysis

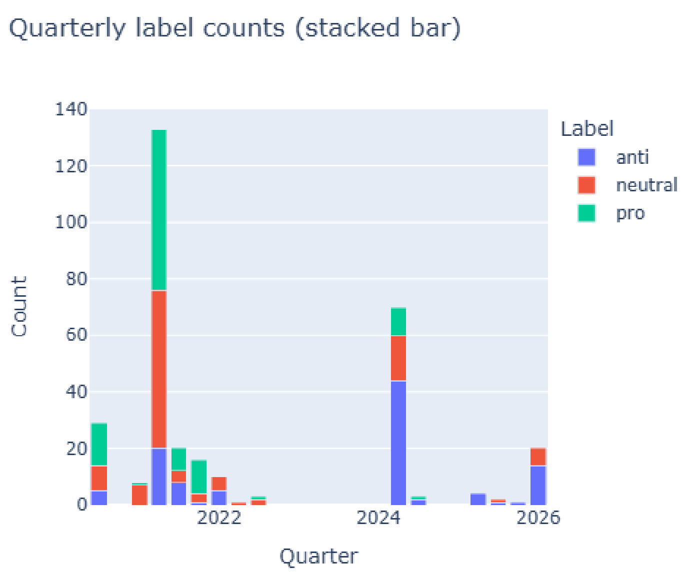

Figure 1. Quarterly distribution of pro, anti, and neutral vaccine-related comments collected from multilingual social media discussions.

Dataset Description

Dataset Characteristics

The final manually labeled dataset consisted of 320 YouTube comments.

Label distribution:

Label	Count
Neutral	110
Anti	105
Pro	105

The dataset was relatively balanced across the three stance categories.

Language Characteristics

The dataset included:

- English comments
- Hindi comments
- Code-mixed English-Hindi comments

Code-mixed social media text introduced additional preprocessing and modeling challenges.

Temporal Characteristics

Date range:

- Start: July 2020
- End: March 2026

The dataset covered 14 unique quarters.

Corpus Properties

The dataset contained noisy social media language including:

- Informal grammar
- Slang
- Misspellings
- Mixed scripts
- Context-dependent statements

These characteristics increased stance classification difficulty.

Temporal Visualization Findings

Temporal analysis revealed:

- Strong discussion spikes during specific quarters
- Uneven temporal distribution of discussions
- Event-driven engagement patterns

Peak stance activity:

Quarter	Label	Count
2021Q2	Neutral	56
2021Q2	Pro	57
2024Q2	Anti	44

The temporal heatmap demonstrated that social media engagement occurred in intermittent bursts rather than continuous discussion activity.

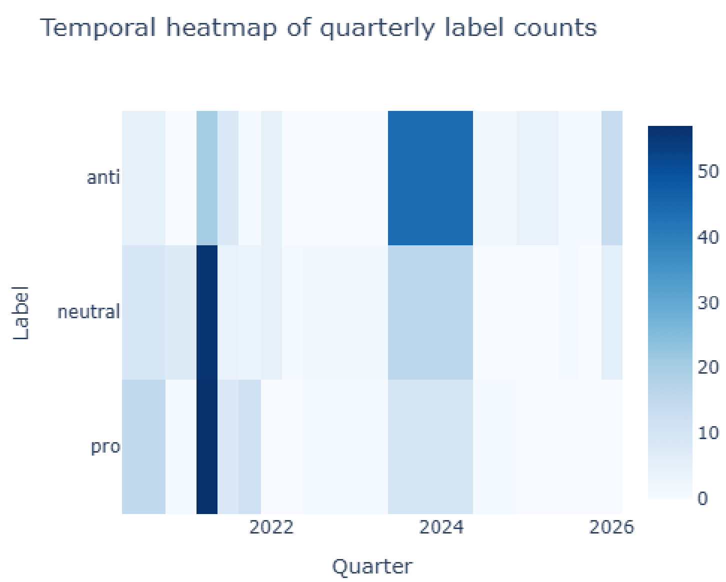

Figure 2. Heatmap showing temporal concentration of stance-related discussions across quarters.

Experimental Setup

Train-Test Split

The dataset was divided into:

- Training set: 80%
- Test set: 20%

An additional validation split was used for deep learning experiments.

Final split sizes:

Split	Size
Train	204
Validation	52
Test	64

Stratified splitting was applied to preserve label balance.

Hardware Environment

Experiments were conducted primarily on CPU.

This limited:

- transformer training duration
- experimentation with larger architectures
- extensive hyperparameter tuning

Preprocessing Pipeline

The preprocessing pipeline included:

- lowercasing
- URL removal
- mention removal
- whitespace normalization
- multilingual character preservation

Balancing Techniques

The following balancing approaches were used:

- `class_weight='balanced'` for LinearSVC
- weighted cross-entropy loss for CNN and mBERT

No oversampling or synthetic data generation techniques were applied.

Evaluation Metrics

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Macro F1-score
- Confusion matrices

Macro F1-score was considered particularly important because it evaluates performance equally across all classes.

Results and Discussion

Model Comparison

Model	Accuracy	Macro F1
TF-IDF + LinearSVC	56.25%	54.92%
CNN	65.62%	62.19%
mBERT	57.81%	57.59%

TF-IDF + LinearSVC Performance

The baseline TF-IDF + LinearSVC model achieved:

- Accuracy: 56.25%
- Macro F1-score: 54.92%

The baseline model successfully learned meaningful stance patterns despite the limited dataset size.

The inclusion of character-level TF-IDF features improved robustness toward:

- spelling variations
- noisy text
- code-mixed language patterns

An important observation was that preserving multilingual characters improved the baseline model performance. Initially, removing non-English characters reduced multilingual information and negatively affected classification quality.

The neutral class remained the most difficult class because neutral comments frequently shared semantic overlap with both pro and anti comments.

CNN Performance

The CNN model produced the best overall performance:

- Accuracy: 65.62%
- Macro F1-score: 62.19%

The CNN outperformed both the baseline model and mBERT.

Possible reasons include:

- CNN effectiveness on short-text classification
- Better learning of local semantic patterns
- Reduced dependency on large-scale datasets compared to transformers
- Better handling of noisy phrase-level information

The CNN strongly identified pro-vaccine comments:

- Pro recall reached approximately 95%

However, the neutral class still remained challenging.

Training logs showed decreasing training loss while validation improvements slowed after several epochs, indicating mild overfitting.

Early stopping helped reduce overfitting and improved generalization.

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mBERT Performance

The multilingual BERT model achieved:

- Accuracy: 57.81%
- Macro F1-score: 57.59%

Although mBERT did not outperform the CNN, its performance remained competitive considering:

- the small dataset size

- multilingual input
- CPU-only training
- noisy social media text

The transformer model showed more balanced precision across classes compared to the CNN.

However, mBERT required more training data to fully utilize its contextual representation capabilities.

Experiments with additional epochs caused overfitting and reduced test performance.

This observation highlights an important limitation of transformer-based models on small datasets.

Neutral Class Difficulty

Across all models, the neutral class consistently produced lower recall and F1-scores.

Reasons include:

- semantic overlap with pro and anti classes
- ambiguous stance expressions
- sarcasm and indirect wording
- short comment length
- noisy social media writing patterns

This observation demonstrates the inherent difficulty of stance classification in real-world social media discussions.

Manual Annotation vs GPT-assisted Labeling

During experimentation, GPT-assisted labeling was also explored.

Although GPT-generated labels increased overall accuracy, they also introduced:

- stronger class imbalance
- reduced macro F1-score
- reduced label diversity

Manual annotation produced more balanced and realistic stance distributions.

Additionally, human annotation considered contextual meaning and nuanced stance expressions that were difficult to reproduce automatically.

This experiment highlighted the importance of careful annotation design in supervised NLP tasks.

Temporal Analysis Discussion

Quarterly temporal analysis revealed several important trends:

- Social media engagement occurred in concentrated bursts.
- Pro and neutral comments peaked during 2021Q2.
- Anti-vaccine comments peaked during 2024Q2.
- Discussion activity declined significantly after major peak periods.

The heatmap visualization effectively highlighted temporal clustering and stance intensity.

Quarterly aggregation proved more appropriate than monthly aggregation because:

- the dataset size was relatively small,
- monthly bins produced unstable statistics,
- quarterly grouping reduced visualization noise.

Overall, the temporal analysis demonstrated that vaccine-related discussions were event-driven rather than continuously active.

Scope for Future Work

Several future improvements can extend this project.

Larger Annotated Dataset

The most important improvement would be increasing the manually labeled dataset size.

A larger dataset would likely improve:

- transformer performance
- class separability
- generalization capability

Semi-supervised Learning

Only 320 comments were manually labeled despite having approximately 3,500 scraped comments.

Future work may explore:

- semi-supervised learning
- pseudo-labeling
- active learning
- self-training approaches

These techniques could leverage the remaining unlabeled data.

Future work may additionally explore unsupervised and self-supervised learning approaches for multilingual stance detection. Topic modeling, clustering, and semantic grouping methods may help identify hidden discussion patterns within unlabeled vaccine-related comments. Self-supervised representation learning and semi-supervised approaches may further improve utilization of the large amount of unlabeled multilingual social media data collected during the study.

True Multimodal Fusion

Future work may revisit multimodal integration using:

- paired image-text datasets
- meme classification
- video metadata
- audio transcripts

More suitable multimodal datasets would enable deeper fusion experiments.

Geospatial Analysis

If reliable geographic metadata becomes available, future work may explore:

- hotspot detection
- regional misinformation analysis
- state-level hesitancy mapping

Larger Transformer Models

Future experiments may evaluate:

- XLM-RoBERTa
- IndicBERT
- domain-adapted multilingual transformers

GPU-based training could significantly improve transformer fine-tuning.

Context-aware Classification

An important limitation of the current work is that models classified comments independently without considering video context.

Future work may incorporate:

- video metadata
- conversation threads
- parent-child comment relationships
- contextual stance modeling

This may improve classification consistency.

Conclusion

This project explored multilingual vaccine hesitancy detection using social media comments collected from YouTube. The study focused on classifying multilingual and code-mixed comments into pro, anti, and neutral stance categories using traditional machine learning, deep learning, and transformer-based approaches.

Three different modeling approaches were implemented: TF-IDF + LinearSVC, CNN, and multilingual BERT (mBERT). Among the evaluated models, the CNN achieved the best overall performance with 65.62% accuracy and a macro F1-score of 62.19%, outperforming both the classical baseline and the transformer-based model on the limited dataset.

The experiments demonstrated that multilingual preprocessing decisions, annotation consistency, and dataset design significantly influence stance classification performance. The study also showed that transformer-based models do not necessarily outperform simpler architectures when training data is limited.

Temporal analysis further revealed that vaccine-related discussions occurred in concentrated periods rather than uniformly over time, with distinct quarterly peaks for different stance categories.

Overall, the project highlighted both the potential and the practical challenges of multilingual social media stance detection in noisy real-world environments. The findings contribute toward understanding how machine learning and NLP techniques may assist future public health misinformation monitoring systems.
